

Question_10:

Code:

```
1  /*
2  Question No: 10
3  There are M rows with different numbers of empty seats.
4  There are N football fans.
5  A ticket from a row with K empty seats costs K pounds.
6  Find the maximum money the club can earn.
7  Always sell from the row having the most empty seats.
8  */
9
10 #include <iostream>
11 #include <queue>
12 using namespace std;
13
14 void heapify(int arr[],int n,int i){
15     int mx=i,l=2*i+1,r=2*i+2;
16
17     if(l<n && arr[l]>arr[mx])
18         mx=l;
19     if(r<n && arr[r]>arr[mx])
20         mx=r;
21     if(mx!=i){
22         swap(arr[i],arr[mx]);
23         heapify(arr,n,mx);
24     }
25 }
26 int main(){
27     int m,n;
28     cin>>m>>n;
29
30     priority_queue<int> q;
31     for(int i=0;i<m;i++){
32         int x;
33         cin>>x;
34         q.push(x);
35     }
36     long long ans=0;
37
38     while(n--){
39         int x=q.top();
40         q.pop();
41         ans+=x;
42         if(x>1)
43             q.push(x-1);
44     }
45     cout<<ans;
46 }
```

Output:

Test Case_1:

```
3 4
1 2 4
11
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
8 4
3 7 1 2 4 7 5 9
31
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```